Section A: MCQs (10 Marks)

(Each question: 1 marks)

Q1 In LogGOPS, if message size increases significantly, which component becomes dominant in communication time? a) Latency (L) b) CPU overhead (o) c) Gap per byte (G) d) Number of processes (P)	Apply
Q2. Which of the following best describes a *communicator* in MPI? a) A memory space shared between processes b) A group of processes with a communication context c) A network protocol d) A scheduling policy	Understand
Q3. Which MPI routine is responsible for ensuring that all processes reach a synchronisation point before proceeding? a) MPI_Finalize b) MPI_Barrier c) MPI_Comm_size d) MPI_Test	Remember
Q4. Recursive doubling is preferred for small messages because a) It maximizes bandwidth b) It reduces number of communication steps c) It avoids synchronisation d) It uses fewer processes	Apply
Q5. What happens when message size exceeds threshold S in LogGOPS? a) Immediate transmission b) Message is dropped c) Rendezvous protocol is used d) Bandwidth becomes infinite	Understand
Q6. In a distributed-memory HPC system, which scenario will most likely degrade performance even if more processors are added? a) High computation-to-communication ratio b) Frequent synchronization with small messages c) Large independent workloads per process d) Embarrassingly parallel tasks	Analyse
Q7. Which MPI function is used to determine the identity of a process? a) MPI_Init	Remember

b) MPI_Comm_size c) MPI_Comm_rank d) MPI_Finalize	
Q8. A parallel application shows no performance improvement beyond 16 processes. Which is the most probable root cause? a) Increased clock speed b) Reduced memory usage c) Dominance of sequential portion and communication overhead d) Compiler inefficiency	Analyse
Q9. In MPI, which situation can lead to idle time even when processes are available? a) Using MPI_Comm_rank b) Load imbalance among processes c) Using collective operations d) High CPU frequency	Understand
Q10. An MPI application performs a reduction operation on a large dataset across many processes. The current implementation uses recursive doubling, and performance is poor. Which of the following is the best modification to improve performance? a) Replace it with MPI_Send/MPI_Recv b) Switch to a ring-based Allreduce algorithm c) Increase the number of processes further d) Use MPI_Barrier before reduction	Apply
Q11. In LogGOPS-based performance modelling, which scenario indicates that latency (L) is the dominant factor? a) Very large messages with continuous streaming b) Frequent small message exchanges between processes c) High computation per process with minimal communication d) Large number of floating-point operations	Analyse

Section B: Short Answer (9 Marks)

(Each question: 3 marks)

Q12. Explain using an example how message size influences the choice of MPI collective algorithms.	Apply
Q13. MPI provides multiple algorithms for collective operations. Evaluate why selecting an appropriate algorithm is critical for performance.	**Evaluate**
Q14. An MPI application exchanges a large number of small messages. Analyse how this communication pattern impacts performance in terms of latency and bandwidth.	Analyse

Section C: Long Answer (10 Marks)

(Each question: 5 marks)

Q15. An MPI-based scientific application running on a cluster shows poor scalability beyond 32 processes. Profiling indicates that the application spends a significant amount of time in frequent small message exchanges and synchronisation points. How can the scalability be improved?	Create
Q16. You are given an MPI-based application where each process performs a portion of computation and periodically exchanges data with other processes. It is observed that: ● When communication is reduced, computation increases significantly ● When computation is reduced, communication becomes very frequent ● Both extremes lead to poor performance Design a theoretical strategy to achieve an optimal balance between communication and computation for this application. Clearly justify your approach using relevant HPC concepts.	Create